



COMPLETE APTITUDE CHEAT SHEET

Master Every Topic: Basic → Advanced

1

NUMBER SYSTEM, HCF & LCM

A. What is This Topic?

Understanding how numbers work, their types, and how to find common factors/multiples—essential for all aptitude exams.

B. Key Concepts

- **Natural Numbers:** 1, 2, 3, 4... (counting numbers)
- **Whole Numbers:** 0, 1, 2, 3... (natural numbers + 0)
- **Prime Numbers:** Only divisible by 1 and itself (2, 3, 5, 7, 11, 13...)
- **Composite Numbers:** Have factors other than 1 and itself
- **HCF (Highest Common Factor):** Largest number that divides two or more numbers
- **LCM (Least Common Multiple):** Smallest number that is divisible by two or more numbers

C. Important Formulas

$$\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b$$

For Three Numbers: Use Prime Factorization Method

Remember: The formula $\text{HCF} \times \text{LCM} = a \times b$ works ONLY for two numbers. For three or more, use prime factorization.

D. Shortcuts & Tricks

- 💧 If numbers are consecutive, $HCF = 1$ (except if one divides the other)
- 💧 If one number is a multiple of another, $HCF = \text{smaller number}$, $LCM = \text{larger number}$
- 💧 For LCM of consecutive numbers: Use prime factorization, it's fastest
- 💧 Prime Factorization Method: Break into primes, then $HCF = \text{lowest powers}$, $LCM = \text{highest powers}$

E. Solved Examples

LEVEL BASIC: Find HCF of 12 and 18

Solution: Factors of 12: 1, 2, 3, 4, 6, 12

Factors of 18: 1, 2, 3, 6, 9, 18

Common factors: 1, 2, 3, 6

Highest = 6

Answer: HCF = 6

LEVEL INTERMEDIATE: Find LCM of 12 and 18

Solution: $12 = 2^2 \times 3$

$18 = 2 \times 3^2$

$LCM = 2^2 \times 3^2 = 4 \times 9 = 36$

Answer: LCM = 36

LEVEL ADVANCED: Three bells ring at 12, 15, 20 minute intervals. When do they ring together?

Solution: Find $LCM(12, 15, 20)$

$12 = 2^2 \times 3$

$15 = 3 \times 5$

$20 = 2^2 \times 5$

$LCM = 2^2 \times 3 \times 5 = 60$

Answer: After 60 minutes (1 hour)

F. Common Mistakes

- ✗ Confusing HCF and LCM

✗ Forgetting that $\text{HCF} \times \text{LCM} = a \times b$ only works for 2 numbers

✗ Not using prime factorization for larger numbers

G. Exam Tips

📍 PLACEMENT: Direct HCF/LCM questions are rare. Usually asked in word problems about time, work, or distribution.

📍 CAT: Time management is crucial. Use prime factorization method—it's fastest.

📍 UPSC: May appear in divisibility and number theory contexts.

H. Practice Questions

1. Find HCF of 48, 64, and 80.
2. Find LCM of 15, 25, and 35.
3. If $\text{HCF}(a,b)=6$ and $\text{LCM}(a,b)=60$, and one number is 12, find the other.
4. Flowers are 24, 36, and 48 in number. Maximum bouquets with equal flowers in each bouquet?

2

SIMPLIFICATION & APPROXIMATION

A. What is This Topic?

Quickly reducing complex expressions and estimating answers—essential for time management in exams.

B. Key Concepts

- **BODMAS Rule:** Brackets → Orders → Division & Multiplication → Addition & Subtraction
- **Approximation:** Rounding to make calculations faster
- **Estimation:** Quick mental calculation without exact answer

- **Algebraic Identities:** $(a+b)^2 = a^2 + 2ab + b^2$, $(a-b)^2 = a^2 - 2ab + b^2$, $(a^2-b^2) = (a+b)(a-b)$

C. Important Rules

BODMAS: Brackets → Orders → Division & Multiplication (L→R) → Addition & Subtraction (L→R)

$$(a+b)(a-b) = a^2 - b^2$$

D. Shortcuts & Tricks

- 💧 For approximate products: Use $(a+b)(a-b) = a^2 - b^2$ formula
- 💧 Example: $102 \times 98 = (100+2)(100-2) = 100^2 - 4 = 10000 - 4 = 9996$
- 💧 Approximation: Round to nearest 5, 10, or 100 for quick mental math
- 💧 Cancellation: Look for common factors before multiplying

E. Solved Examples

LEVEL BASIC: Simplify $5 + 3 \times 2$

Solution: Follow BODMAS.

Multiply first: $3 \times 2 = 6$

Then add: $5 + 6 = 11$

Answer: 11

LEVEL INTERMEDIATE: Simplify $(12 \times 8) / (4 \times 2) + 5$

Solution: Brackets first: $(12 \times 8) = 96$, $(4 \times 2) = 8$

Division: $96 / 8 = 12$

Addition: $12 + 5 = 17$

Answer: 17

LEVEL ADVANCED: Approximate 102×98 without actual multiplication

Solution: Use $(a+b)(a-b) = a^2 - b^2$

Here $a = 100$, $b = 2$

$(100+2)(100-2) = 100^2 - 2^2 = 10000 - 4 = 9996$

Answer: 9996

F. Common Mistakes

- ✗ Doing addition before multiplication (violating BODMAS)
- ✗ Over-approximating and losing accuracy needed for exam
- ✗ Forgetting bracket operations must be done FIRST

G. Exam Tips

- 📍 PLACEMENT: Use approximation to save time—answers usually have wide differences.
- 📍 CAT: Exact answer matters more; approximate only when confident.
- 📍 UPSC: Show your workings; approximation must be justified.

H. Practice Questions

1. Simplify: $8 \div 2 + 5 \times 3 - 1$
2. Approximate: 104×96 (without calculator)
3. Simplify: $(25 \times 12) / (5 \times 2) + 4$

3 PERCENTAGE

A. What is This Topic?

Understanding ratios as parts of 100. The foundation for profit, loss, interest, and data interpretation.

B. Key Concepts

- **Percentage:** Value expressed as a part of 100

- **Formula:** $\text{Percentage} = (\text{Part} / \text{Whole}) \times 100$
- **Reverse Calculation:** If $x\%$ of a number is y , then $\text{number} = (y \times 100) / x$
- **Percentage Change:** $[(\text{New} - \text{Old}) / \text{Old}] \times 100$

C. Important Formulas

$$x\% \text{ of } y = (x/100) \times y$$

$$\% \text{ Change} = [(\text{Final} - \text{Initial}) / \text{Initial}] \times 100$$

$$\text{Successive \% Change} = (100 + a)(100 + b) / 100$$

D. Shortcuts & Tricks

💧 Common conversions: $25\% = 1/4$, $50\% = 1/2$, $75\% = 3/4$, $33.33\% = 1/3$, $66.67\% = 2/3$, $20\% = 1/5$

💧 Successive changes: $+20\%$ then $-20\% \neq 0\%$. It's actually $(120 \times 80)/100 = 96 = -4\%$ net

💧 Percentage points vs percentage: 5% increase is different from 5 percentage points

E. Solved Examples

LEVEL BASIC: Find 25% of 80

Solution: $25\% \text{ of } 80 = (25/100) \times 80 = 0.25 \times 80 = 20$

Answer: 20

LEVEL INTERMEDIATE: If 40% of $x = 120$, find x

Solution: $0.4x = 120$

$x = 120 / 0.4 = 300$

Answer: $x = 300$

LEVEL ADVANCED: Price increases by 20%, then decreases by 20%. Net change?

Solution: Assume original = 100

After 20% increase: $100 \times 1.2 = 120$

After 20% decrease: $120 \times 0.8 = 96$

Net change: $96 - 100 = -4$

% change = -4%

Answer: -4% (a loss, not 0%)

F. Common Mistakes

- ✗ Forgetting to convert percentage to decimal/fraction
- ✗ Assuming +20% then -20% = 0% (it's actually -4%)
- ✗ Mixing percentage and percentage points

G. Exam Tips

- 📍 PLACEMENT: Very common in salary, discount, and growth questions. Practice mental math.
- 📍 CAT: Often combined with other topics (profit, loss, interest).
- 📍 UPSC: Appears in data interpretation questions frequently.

H. Practice Questions

- Find 15% of 200.
- If 30% of a number is 90, what is 40% of that number?
- Price goes from Rs 100 to Rs 120. What is the percentage increase?

4 PROFIT AND LOSS

A. What is This Topic?

Buying and selling transactions with profit/loss calculations—heavily tested in all competitive exams.

B. Key Concepts

- **Cost Price (CP):** Amount paid to buy an item
- **Selling Price (SP):** Amount received on selling
- **Marked Price (MP):** Price displayed on item
- **Profit:** $SP > CP \rightarrow \text{Profit} = SP - CP$
- **Loss:** $SP < CP \rightarrow \text{Loss} = CP - SP$
- **Discount:** Reduction on Marked Price

C. Important Formulas

$$\text{Profit \%} = (\text{Profit} / \text{CP}) \times 100 = ((\text{SP} - \text{CP}) / \text{CP}) \times 100$$

$$\text{Loss \%} = (\text{Loss} / \text{CP}) \times 100 = ((\text{CP} - \text{SP}) / \text{CP}) \times 100$$

$$\text{SP} = \text{CP} \times (1 + \text{Profit\%/100})$$

$$\text{SP} = \text{CP} \times (1 - \text{Loss\%/100})$$

$$\text{SP} = \text{MP} - \text{Discount}$$

D. Shortcuts & Tricks

- 💧 Mark Up is on CP, but Discount is on MP—don't confuse them!
- 💧 When two items sold at same SP with different profit %: Calculate each separately, then combine
- 💧 Profit % is ALWAYS calculated on CP, not SP
- 💧 For quick calculation: If $CP=100$, then $SP = 100 + \text{Profit\%}$ (easier to visualize)

E. Solved Examples

LEVEL BASIC: Bought at Rs 100, sold at Rs 120. Profit %?

Solution: Profit = 120 - 100 = Rs 20

Profit% = $(20/100) \times 100 = 20\%$

Answer: 20%

LEVEL INTERMEDIATE: Bought at Rs 500. Sold at 25% loss. Selling price?

Solution: Loss% = 25%

SP = $CP(1 - \text{Loss\%/100}) = 500(1 - 0.25) = 500 \times 0.75 = \text{Rs } 375$

Answer: Rs 375

LEVEL ADVANCED: MP Rs 1000, Discount 20%. If profit is 25%, find CP.

Solution: SP = MP - Discount = 1000 - 200 = Rs 800

If Profit 25%, then SP = $CP(1.25)$

$800 = CP \times 1.25$

$CP = 800/1.25 = \text{Rs } 640$

Answer: Rs 640

F. Common Mistakes

- ✗ Calculating profit % on SP instead of CP
- ✗ Confusing Mark Price with Cost Price
- ✗ Forgetting discount is applied on MP, not CP

G. Exam Tips

- 🕒 PLACEMENT: Very common question type. Practice until automatic.
- 🕒 CAT: Often combined with cost price, discount, and shopkeeper profit scenarios.
- 🕒 UPSC: May appear in economic contexts and policy analysis.

H. Practice Questions

1. Bought Rs 200, sold Rs 240. Find profit percent.
2. MP Rs 500, discount 15%. If cost was Rs 350, find profit percent.
3. 2 items: One sold at 20% profit, other at 20% loss, same SP each. Net profit/loss?

5 RATIO AND PROPORTION

A. What is This Topic?

Comparing quantities and finding relationships—essential for mixtures, speeds, work problems, and partnerships.

B. Key Concepts

- **Ratio:** Comparison of two quantities ($a:b = a/b$)
- **Proportion:** Two ratios are equal ($a:b = c:d$)
- **Inverse Ratio:** If $a:b$, then inverse is $b:a$
- **Compound Ratio:** $(a:b) \times (c:d) = ac:bd$

C. Important Formulas

If $a:b = c:d$, then $ad = bc$ (product of extremes = product of means)

If dividing total P in ratio $a:b$, then parts = $(Pa/(a+b))$ and $(Pb/(a+b))$

If $a:b:c$ and total is T , each part = $T \times (\text{component ratio}) / (\text{sum of ratios})$

D. Shortcuts & Tricks

- 💧 To combine ratios with common element: Equalize the common element first
- 💧 Example: $A:B = 2:3$, $B:C = 3:4 \rightarrow A:B:C = 2:3:4$ (B already same)

💧 When simplifying ratios: Find GCD and divide both by it

E. Solved Examples

LEVEL BASIC: Ratio of boys to girls is 3:2. If boys = 12, find girls.

Solution: 3:2 means 3 parts boys, 2 parts girls

If 3 parts = 12, then 1 part = 4

Girls = 2 parts = $2 \times 4 = 8$

Answer: 8 girls

LEVEL INTERMEDIATE: Divide Rs 500 in ratio 2:3:5

Solution: Total parts = $2+3+5 = 10$

Each part = $500/10 = \text{Rs } 50$

Parts = Rs 100, Rs 150, Rs 250

Answer: Rs 100, Rs 150, Rs 250

LEVEL ADVANCED: A:B = 2:3, B:C = 3:4. Find A:B:C

Solution: B is common (=3) in both ratios

First ratio: A:B = 2:3

Second ratio: B:C = 3:4

Since B = 3 in both, A:B:C = 2:3:4

Answer: 2:3:4

F. Common Mistakes

- ✗ Not simplifying ratios to lowest terms
- ✗ Forgetting to add ratio components when dividing totals
- ✗ Confusing a:b with $a/(a+b)$ as a fraction

G. Exam Tips

- 📍 PLACEMENT: Very common in partnerships, mixtures, speed problems.
- 📍 CAT: Often complex multi-ratio problems requiring careful combination.

📌 UPSC: May appear in distribution and allocation contexts.

H. Practice Questions

1. If $A:B = 4:5$ and $B:C = 3:2$, find $A:B:C$.
2. Divide Rs 1000 between X and Y in ratio 3:7.
3. If 5 oranges cost Rs 25, what is the ratio of oranges to cost?

6 AVERAGE

A. What is This Topic?

Finding the central value of a dataset—foundation for statistics and data interpretation questions.

B. Key Concepts

- **Average (Mean):** Sum of all values / Number of values
- **Weighted Average:** $(\sum(\text{value} \times \text{weight})) / (\sum \text{weight})$
- **Range:** Difference between highest and lowest value

C. Important Formulas

$$\text{Average} = (\text{Sum of all values}) / (\text{Number of values})$$

$$\text{New Average} = (\text{Old Sum} \pm \text{Changes}) / (\text{Old Count} \pm \text{New/Removed Members})$$

$$\text{Weighted Average} = (n_1 \times a_1 + n_2 \times a_2) / (n_1 + n_2)$$

D. Shortcuts & Tricks

- 💧 If all values increase by k , average also increases by k
- 💧 If person of age x leaves: New average = $(\text{Old avg} \times n - x) / (n-1)$
- 💧 Weighted average with two groups: $(n_1a_1 + n_2a_2)/(n_1+n_2)$

E. Solved Examples

LEVEL BASIC: Average of 10, 20, 30, 40

Solution: Sum = $10+20+30+40 = 100$

Count = 4

Average = $100/4 = 25$

Answer: 25

LEVEL INTERMEDIATE: Average of 6 numbers is 30. If one more number 42 is added, new average?

Solution: Old sum = $30 \times 6 = 180$

New sum = $180 + 42 = 222$

New count = 7

New avg = $222/7 \approx 31.7$

Answer: 31.7 (or 222/7)

LEVEL ADVANCED: Average age of 5 people is 40. When 25-year-old leaves, new average is 44. Who joined?

Solution: Old sum = $40 \times 5 = 200$

After 25 leaves: $200 - 25 = 175$, count = 4, avg = 43.75

But new avg = 44 with someone new: $44 \times 5 = 220$

Person added = $220 - 175 = 45$

Answer: 45-year-old joined

F. Common Mistakes

- ✗ Using wrong denominator (count vs sum)
- ✗ Forgetting count changes when members leave/join
- ✗ Not recognizing weighted average in two-group problems

G. Exam Tips

- 📍 PLACEMENT: Common in HR data and salary questions.
- 📍 CAT: Usually combined with percentage or other topics.
- 📍 UPSC: May appear in statistical contexts and data analysis.

H. Practice Questions

1. Average of 4 numbers is 25. If one number is 31, find sum of other 3.
2. Average age of 10 students is 15. When teacher (age 40) joins, new average?

7 TIME, SPEED & DISTANCE

A. What is This Topic?

Motion problems from trains to swimming. Fundamental formula with many applications across all exams.

B. Key Concepts

- **Distance = Speed × Time**
- **Relative Speed (same direction):** $|\text{Speed}_1 - \text{Speed}_2|$
- **Relative Speed (opposite direction):** $\text{Speed}_1 + \text{Speed}_2$
- **Unit Conversion:** $1 \text{ m/s} = 3.6 \text{ km/h}$

C. Important Formulas

$$\text{Distance} = \text{Speed} \times \text{Time} \quad | \quad \text{Speed} = \text{Distance} / \text{Time} \quad | \quad \text{Time} = \text{Distance} / \text{Speed}$$

$$\text{Average Speed} = \text{Total Distance} / \text{Total Time} \quad (\text{NOT average of speeds})$$

When distance is same: Avg Speed = $2S_1S_2/(S_1+S_2)$ [Harmonic Mean]

When time is same: Avg Speed = $(S_1+S_2)/2$ [Arithmetic Mean]

D. Shortcuts & Tricks

💧 If distance same: Avg speed = $(2 \times S_1 \times S_2)/(S_1 + S_2)$ [harmonic mean]

💧 If time same: Avg speed = $(S_1 + S_2)/2$ [arithmetic mean]

💧 For train crossing: Total distance = Train length + Object length

💧 km/h to m/s: Multiply by 5/18. m/s to km/h: Multiply by 18/5

E. Solved Examples

LEVEL BASIC: Car travels 150 km in 3 hours. Speed?

Solution: Speed = Distance/Time = $150/3 = 50$ km/h

Answer: 50 km/h

LEVEL INTERMEDIATE: Train A (60 km/h) and Train B (40 km/h) moving toward each other. Relative speed?

Solution: Opposite direction

Relative speed = $60 + 40 = 100$ km/h

Answer: 100 km/h

LEVEL ADVANCED: Person travels 100 km at 50 km/h, then 100 km at 100 km/h. Average speed?

Solution: Time₁ = $100/50 = 2$ hrs

Time₂ = $100/100 = 1$ hr

Total distance = 200 km

Total time = 3 hrs

Avg speed = $200/3 \approx 66.7$ km/h

Answer: 66.7 km/h (or 200/3)

F. Common Mistakes

- ✗ Using average of speeds instead of total distance/total time
- ✗ Forgetting relative speed changes with direction
- ✗ Wrong unit conversions ($\text{km/h} \leftrightarrow \text{m/s}$)

G. Exam Tips

- 📍 PLACEMENT: Very common. Practice unit conversions thoroughly.
- 📍 CAT: Complex multi-part journeys and relative motion problems.
- 📍 UPSC: Less common but appear in logistics and infrastructure contexts.

H. Practice Questions

1. Train 100m long crosses platform in 20s at 54 km/h. Platform length?
2. Person travels 60 km at 30 km/h and 90 km at 60 km/h. Average speed?



TIME & WORK

A. What is This Topic?

How long it takes to complete work—fundamental for productivity, efficiency, and project management problems.

B. Key Concepts

- **Work = Rate $\times$ Time**
- **Rate = $1/\text{Time taken}$** (work per unit time)
- **Combined Rate = Sum of individual rates**

- **Inverse Relationship:** More people = Less time

C. Important Formulas

$$\text{Work Done} = \text{Rate} \times \text{Time} \mid \text{Rate} = 1/\text{Time}$$

$$\text{If A does work in } x \text{ days, B in } y \text{ days: Together} = xy/(x+y) \text{ days}$$

$$\text{Combined Rate} = \text{Rate}_A + \text{Rate}_B + \text{Rate}_C + \dots$$

D. Shortcuts & Tricks

- 💧 Think in terms of WORK RATE (fraction of work per day) not time
- 💧 If A does $1/6$ per day, in 3 days A completes $3/6 = 1/2$ work
- 💧 Combined rate adds: If $A=1/x$, $B=1/y$, then $A+B=1/x+1/y$
- 💧 For two people: If time = $1/(1/x + 1/y) = xy/(x+y)$

E. Solved Examples

LEVEL BASIC: A does work in 10 days. Rate of A?

Solution: Rate = $1/\text{Time} = 1/10$ work per day

Answer: $1/10$ per day

LEVEL INTERMEDIATE: A does work in 6 days, B in 8 days. Together?

Solution: A's rate = $1/6$, B's rate = $1/8$

Combined = $1/6 + 1/8 = 4/24 + 3/24 = 7/24$

Time = $24/7 \approx 3.43$ days

Answer: $24/7$ days ≈ 3.43 days

LEVEL ADVANCED: A, B, C do work in 6, 8, 12 days. A works 2 days, B works 3 days, C completes rest. Time for C?

Solution: A's 2 days = $2 \times 1/6 = 1/3$ work

B's 3 days = $3 \times 1/8 = 3/8$ work

$$\text{Total} = 1/3 + 3/8 = 8/24 + 9/24 = 17/24$$

$$\text{Remaining} = 7/24$$

$$\text{C's rate} = 1/12 = 2/24 \text{ per day}$$

$$\text{C needs} = (7/24) / (2/24) = 3.5 \text{ days}$$

Answer: 3.5 days or 7/2 days

9 PIPES & CISTERNS

A. What is This Topic?

Filling and emptying tanks/pools using pipes with different rates—extension of time & work concept.

B. Key Concepts

- **Inlet Pipe:** Fills the tank (positive work rate)
- **Outlet Pipe:** Empties the tank (negative work rate)
- **Net Rate:** Inlet rate - Outlet rate
- **Capacity:** Total volume of tank

C. Important Formulas

$$\text{Rate} = \text{Capacity} / \text{Time} \mid \text{Time} = \text{Capacity} / \text{Rate}$$

$$\text{Combined Rate} = \text{Rate}_1 + \text{Rate}_2 + \dots - \text{Outlet Rates}$$

$$\text{If Inlet fills in } x \text{ hours, Outlet empties in } y: \text{ Together} = xy/(y-x) \text{ [when } x < y]$$

D. Shortcuts & Tricks

💧 Think of rates as "fraction of tank filled per hour"

💧 Outlet rate is NEGATIVE when combined with inlet rates

💧 If both open simultaneously: Add inlet, subtract outlet rates

E. Solved Examples

LEVEL BASIC: Pipe A fills tank in 6 hours. Fraction filled in 2 hours?

Solution: Rate = $1/6$ per hour

In 2 hours: $2 \times (1/6) = 2/6 = 1/3$

Answer: $1/3$ tank

LEVEL INTERMEDIATE: Pipe A fills in 8 hours, Pipe B empties in 12 hours. Both open. Time to fill?

Solution: A's rate = $1/8$ (positive)

B's rate = $-1/12$ (negative)

Net = $1/8 - 1/12 = 3/24 - 2/24 = 1/24$

Time = 24 hours

Answer: 24 hours

LEVEL ADVANCED: A fills in 10, B fills in 15, C empties in 20. All three open together. How long to fill?

Solution: A's rate = $1/10$

B's rate = $1/15$

C's rate = $-1/20$

Combined = $1/10 + 1/15 - 1/20 = 6/60 + 4/60 - 3/60 = 7/60$

Time = $60/7 \approx 8.57$ hours

Answer: $60/7$ hours ≈ 8.57 hours

F. Common Mistakes

- ✗ Adding outlet rate instead of subtracting
- ✗ Confusing "fill time" with "drain time" directions
- ✗ Not converting rates to same denominator

G. Exam Tips

📍 PLACEMENT: Direct pipe problems with 2-3 pipes

- 🕒 CAT: Complex scenarios with varying rates or partial filling
- 🕒 UPSC: Less common, but appear in water resource management

H. Practice Questions

1. Pipe A fills tank in 5 hours, B fills in 6 hours. Together?
2. Tank filled in 20 hours by pipe A alone. If drain opens, time becomes 30 hours. Drain time?
3. Two pipes fill in 12 and 15 hours. Drain empties in 20 hours. All open. Time to fill?

10 PROBLEMS ON AGES

A. What is This Topic?

Finding relationships between ages of different people at different points in time—uses ratios and linear equations.

B. Key Concepts

- **Current Age:** Age right now
- **Age n years ago:** Current age - n
- **Age n years later:** Current age + n
- **Age Ratio:** Relationship between two people's ages

C. Important Formulas

If ratio $a:b$, then ages are ax and bx (where x is multiplier)

Age difference is constant: If diff today = k , then diff n years ago = k , diff n years later = k

D. Shortcuts & Tricks

- 💧 Age difference NEVER changes over time
- 💧 If ratio is a:b at one time, it doesn't stay same at other times
- 💧 Use x as multiplier: ages are ax, bx, then form equations

E. Solved Examples

LEVEL BASIC: Ratio of A's age to B's age is 4:3. A is 24. Find B's age.

Solution: 4:3 means $4x$ and $3x$

$$4x = 24 \rightarrow x = 6$$

$$\text{B's age} = 3x = 3(6) = 18$$

Answer: 18 years

LEVEL INTERMEDIATE: Father is 40, son is 10. In how many years will father be 3 times the son's age?

Solution: Let x = years later

$$\text{Father's age} = 40 + x$$

$$\text{Son's age} = 10 + x$$

$$\text{Condition: } 40 + x = 3(10 + x)$$

$$40 + x = 30 + 3x$$

$$10 = 2x \rightarrow x = 5$$

Answer: 5 years

LEVEL ADVANCED: Present ratio A:B = 3:2. 5 years ago, ratio was 5:3. Find current ages.

Solution: Current: $A = 3x$, $B = 2x$

5 years ago: A was $3x-5$, B was $2x-5$

$$\text{Ratio: } (3x-5):(2x-5) = 5:3$$

$$(3x-5)/(2x-5) = 5/3$$

$$3(3x-5) = 5(2x-5)$$

$$9x - 15 = 10x - 25$$

$$x = 10$$

$$A = 30, B = 20$$

Answer: A = 30, B = 20 years

F. Common Mistakes

- ✗ Thinking age ratio stays same over time
- ✗ Forgetting to set up equation correctly with variables
- ✗ Not using the fact that age difference is constant

G. Exam Tips

- 📍 PLACEMENT: 2-person age problems, simple ratios
- 📍 CAT: Complex multi-person scenarios, past and future times
- 📍 UPSC: May appear in population/demographics contexts

H. Practice Questions

1. A is 5 years older than B. 4 years ago, A was twice B's age. Find current ages.
2. Ratio of ages of A and B is 5:4. 3 years later, ratio becomes 11:9. Find A's age.
3. Father's age is sum of 3 children. 10 years later, father's age = sum of children. Ages?

1 1 PARTNERSHIP

A. What is This Topic?

Dividing profit/loss among partners based on capital invested and time period—uses ratio and proportion.

B. Key Concepts

- **Capital:** Amount invested by partner
- **Time:** Duration for which capital is invested

- **Profit Ratio:** Usually proportional to (Capital × Time)
- **Simple Partnership:** All invest for same time
- **Compound Partnership:** Different investment times

C. Important Formulas

$$\text{Profit Ratio} = (\text{Capital}_1 \times \text{Time}_1) : (\text{Capital}_2 \times \text{Time}_2)$$

$$\text{If same time period: Profit Ratio} = \text{Capital Ratio}$$

$$\text{If same capital: Profit Ratio} = \text{Time Ratio}$$

D. Shortcuts & Tricks

- 💧 Profit is shared in ratio of (Capital × Time)
- 💧 If one partner joins/leaves mid-year: Count only their months
- 💧 Use Capital × Time as numerator for each partner, then simplify ratio

E. Solved Examples

LEVEL BASIC: A invests Rs 10000, B invests Rs 15000. Profit is Rs 500. How to divide?

Solution: Both invest for same time

$$\text{Profit Ratio} = \text{Capital Ratio} = 10000:15000 = 2:3$$

$$\text{A gets} = (2/5) \times 500 = \text{Rs } 200$$

$$\text{B gets} = (3/5) \times 500 = \text{Rs } 300$$

Answer: A = Rs 200, B = Rs 300

LEVEL INTERMEDIATE: A invests Rs 20000 for 12 months, B invests Rs 30000 for 8 months. Profit Rs 1000. Share?

$$\text{Solution: A's investment} = 20000 \times 12 = 240000$$

$$\text{B's investment} = 30000 \times 8 = 240000$$

$$\text{Ratio} = 240000:240000 = 1:1$$

$$\text{Each gets} = \text{Rs } 500$$

Answer: A = Rs 500, B = Rs 500

LEVEL ADVANCED: A invests Rs 15000 for 12 months, B invests Rs 20000 for 6 months, C invests Rs 25000 for 9 months. Total profit Rs 3000. Share?

Solution: $A = 15000 \times 12 = 180000$

$B = 20000 \times 6 = 120000$

$C = 25000 \times 9 = 225000$

Ratio = $180000:120000:225000 = 12:8:15$

Total = 35 parts

$A = (12/35) \times 3000 \approx \text{Rs } 1029$

$B = (8/35) \times 3000 \approx \text{Rs } 686$

$C = (15/35) \times 3000 \approx \text{Rs } 1286$

Answer: A $\approx$ Rs 1029, B $\approx$ Rs 686, C $\approx$ Rs 1286

F. Common Mistakes

- ✗ Forgetting to multiply capital by time
- ✗ Using only capital or only time, not both
- ✗ Not counting months correctly for partial year investment

G. Exam Tips

- 🕒 PLACEMENT: Simple 2-3 partner problems with clear numbers
- 🕒 CAT: Complex multi-partner scenarios with varying times
- 🕒 UPSC: May appear in economic partnership contexts

H. Practice Questions

1. A and B enter partnership with Rs 50000 and Rs 60000. After 2 years, profit is Rs 4400. Find A's share.
2. X invests Rs 2000 for 6 months, Y invests Rs 3000 for 4 months. Profit Rs 600. Divide profit.

1 2 BOATS & STREAMS

A. What is This Topic?

Speed problems when boat moves in water with current—extension of relative speed concept.

B. Key Concepts

- **Boat Speed (B):** Speed of boat in still water
- **Current Speed (C):** Speed of water flow
- **Downstream:** Moving with current, speed = $B + C$
- **Upstream:** Moving against current, speed = $B - C$

C. Important Formulas

$$\text{Downstream Speed} = \text{Boat Speed} + \text{Current Speed} (B + C)$$

$$\text{Upstream Speed} = \text{Boat Speed} - \text{Current Speed} (B - C)$$

$$\text{Boat Speed} = (\text{Downstream} + \text{Upstream}) / 2$$

$$\text{Current Speed} = (\text{Downstream} - \text{Upstream}) / 2$$

D. Shortcuts & Tricks

- 💧 Average of downstream and upstream speeds = Boat speed
- 💧 Half the difference of downstream and upstream = Current speed
- 💧 Time downstream < Time upstream (for same distance)

E. Solved Examples

LEVEL BASIC: Boat speed = 10 km/h, Current = 2 km/h. Downstream and upstream speeds?

Solution: Downstream = $10 + 2 = 12$ km/h

Upstream = $10 - 2 = 8$ km/h

Answer: D = 12 km/h, U = 8 km/h

LEVEL INTERMEDIATE: Boat goes 48 km downstream in 3 hours, 48 km upstream in 4 hours. Find boat and current speed.

Solution: Downstream speed = $48/3 = 16$ km/h

Upstream speed = $48/4 = 12$ km/h

Boat speed = $(16 + 12)/2 = 14$ km/h

Current = $(16 - 12)/2 = 2$ km/h

Answer: Boat = 14 km/h, Current = 2 km/h

LEVEL ADVANCED: Boat takes 5 hours downstream, 7 hours upstream for same distance. Current is 2 km/h. Find boat speed and distance.

Solution: Let boat speed = b , distance = d

Downstream: $d/(b+2) = 5 \rightarrow d = 5(b+2)$

Upstream: $d/(b-2) = 7 \rightarrow d = 7(b-2)$

$$5(b+2) = 7(b-2)$$

$$5b + 10 = 7b - 14$$

$$24 = 2b \rightarrow b = 12 \text{ km/h}$$

$$d = 5(12+2) = 70 \text{ km}$$

Answer: Boat = 12 km/h, Distance = 70 km

F. Common Mistakes

- ✗ Adding current when going upstream (should subtract)
- ✗ Confusing which speed is faster (downstream is faster)
- ✗ Using same time for both upstream and downstream (time is different)

G. Exam Tips

- 📍 PLACEMENT: Basic downstream/upstream with given speeds

📍 CAT: Complex scenarios with variable speed, roundtrips

📍 UPSC: May appear in transportation/logistics contexts

H. Practice Questions

1. Boat speed 12 km/h, current 3 km/h. Time to cover 60 km downstream and 40 km upstream?
2. A boat covers 100 km downstream in 5 hours and upstream in 10 hours. Find current speed.

1 3 SIMPLE & COMPOUND INTEREST

A. What is This Topic?

Money growth calculations—Simple Interest on principal only, Compound Interest on accumulated amount.

B. Key Concepts

- **Principal (P):** Original amount invested
- **Rate (R):** Interest percentage per annum
- **Time (T):** Duration in years
- **Simple Interest:** Fixed interest on principal
- **Compound Interest:** Interest on principal + accumulated interest

C. Important Formulas

$$\text{Simple Interest: } SI = (P \times R \times T) / 100$$

$$\text{Compound Interest: } A = P(1 + R/100)^T, CI = A - P$$

$$\text{Amount in SI: } A = P + SI = P(1 + RT/100)$$

$$CI - SI \text{ (2 years): } P \times R^2 / 10000$$

D. Shortcuts & Tricks

- 💧 For compound interest: Use $(1 + r/100)^n$, don't expand if $n > 2$
- 💧 If amount becomes n times: $n = (1 + r/100)^t$, solve using logs if needed
- 💧 Half-yearly: Use $R/2$ and $2T$. Quarterly: Use $R/4$ and $4T$
- 💧 $CI - SI$ for 3 years $= P \times R \times (300 + R^2) / 1000000$

E. Solved Examples

LEVEL BASIC: Simple Interest on Rs 1000 at 5% p.a. for 2 years

Solution: $SI = (1000 \times 5 \times 2) / 100 = 10000/100 = \text{Rs } 100$

Answer: Rs 100

LEVEL INTERMEDIATE: Compound Interest on Rs 1000 at 10% p.a. for 2 years

Solution: $A = 1000(1 + 10/100)^2 = 1000(1.1)^2 = 1000 \times 1.21 = \text{Rs } 1210$

$CI = 1210 - 1000 = \text{Rs } 210$

Answer: Rs 210

LEVEL ADVANCED: Rs 2000 becomes Rs 2662 in 2 years at compound interest. Rate?

Solution: $A = P(1 + R/100)^T$

$$2662 = 2000(1 + R/100)^2$$

$$2662/2000 = (1 + R/100)^2$$

$$1.331 = (1 + R/100)^2$$

$$\sqrt{1.331} \approx 1.1535 = 1 + R/100$$

$$R \approx 15.35\% \approx 15\% \text{ (approximately)}$$

Answer: $\approx 15\%$ p.a.

F. Common Mistakes

- ✗ Using simple interest formula for compound interest
- ✗ Forgetting to convert rate and time to same units
- ✗ Not using correct compounding period

G. Exam Tips

- 📍 PLACEMENT: Direct SI and CI calculations
- 📍 CAT: Comparison problems, rate/time finding problems
- 📍 UPSC: Often in economic policy and banking contexts

H. Practice Questions

1. SI on Rs 5000 at 8% p.a. for 3 years?
2. CI on Rs 2000 at 5% p.a. for 2 years?
3. Amount becomes 2 times in 5 years at CI. Rate?

1 4 PERMUTATION & COMBINATION

A. What is This Topic?

Advanced arranging and selecting with constraints—foundation for probability and counting complex scenarios.

B. Key Concepts

- **Permutation:** Arrangements where order matters
- **Combination:** Selection where order doesn't matter
- **Circular Permutation:** Arrangement in a circle
- **Constraints:** With/without repetition, with restrictions

C. Important Formulas

$$P(n,r) = n! / (n-r)! \text{ [Permutation]}$$

$$C(n,r) = n! / (r!(n-r)!) \text{ [Combination]}$$

$$\text{Circular Permutation} = (n-1)! \text{ [arrangements in circle]}$$

$$\text{With repetition: } n^r \text{ [n choices, r positions]}$$

D. Shortcuts & Tricks

💧 $C(n,r) = C(n, n-r)$ [Use smaller value for faster calculation]

💧 $P(n,r) = C(n,r) \times r!$ [Conversion between P and C]

💧 $nCr + nCr-1 = (n+1)Cr$ [Pascal's triangle property]

E. Solved Examples

LEVEL BASIC: How many 3-digit numbers using 1,2,3,4,5 without repetition?

Solution: Need to arrange 3 digits from 5

$$P(5,3) = 5!/(5-3)! = 5!/2! = (5 \times 4 \times 3) = 60$$

Answer: 60 numbers

LEVEL INTERMEDIATE: How many ways to select 3 students from 10 for a committee?

Solution: Order doesn't matter (committee)

$$C(10,3) = 10!/(3! \times 7!) = (10 \times 9 \times 8)/(3 \times 2 \times 1) = 720/6 = 120$$

Answer: 120 ways

LEVEL ADVANCED: 5 people sit around a circular table. How many arrangements (considering rotation as same)?

Solution: Circular permutation = $(n-1)!$

$$= (5-1)! = 4! = 24$$

Answer: 24 arrangements

F. Common Mistakes

- ✗ Using P when C needed (order doesn't matter)
- ✗ Forgetting $0! = 1$
- ✗ Not accounting for constraints or restrictions

G. Exam Tips

- 📍 PLACEMENT: Basic P&C, no complex constraints
- 📍 CAT: Complex constraint problems, advanced counting
- 📍 UPSC: Less common but appear in scheduling problems

H. Practice Questions

1. How many ways to arrange 4 boys and 3 girls in a row such that girls are together?
2. How many 4-digit numbers ≥ 5000 using 1,2,3,4,5,6 without repetition?
3. From 8 people, how many 3-person committees can be formed with 1 specific person always included?

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PROBABILITY

A. What is This Topic?

Advanced probability—conditional probability, independent events, and combined outcomes.

B. Key Concepts

- **Probability:** Measure of likelihood (0 to 1)
- **Sample Space:** All possible outcomes
- **Event:** Subset of sample space
- **Conditional Probability:** Probability given another event occurred
- **Independent Events:** Occurrence of one doesn't affect other

C. Important Formulas

$$P(E) = n(E) / n(S) \text{ [Basic probability]}$$

$$P(A \text{ and } B) = P(A) \times P(B) \text{ [If independent]}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) \text{ [Addition rule]}$$

$$P(A|B) = P(A \text{ and } B) / P(B) \text{ [Conditional probability]}$$

D. Shortcuts & Tricks

💧 $P(\text{not } E) = 1 - P(E)$ [Complement rule]

💧 For independent events: multiply individual probabilities

💧 For mutually exclusive: add probabilities

E. Solved Examples

LEVEL BASIC: Probability of rolling 3 or 4 on a die?

Solution: Favorable outcomes: 3, 4 (2 outcomes)

Total outcomes: 1,2,3,4,5,6 (6 outcomes)

$$P = 2/6 = 1/3$$

Answer: 1/3

LEVEL INTERMEDIATE: Two coins tossed. Probability of at least one head?

Solution: Sample space: HH, HT, TH, TT

At least one head: HH, HT, TH (3 outcomes)

$$P = 3/4$$

$$\text{OR: } P(\text{at least 1 head}) = 1 - P(\text{no head}) = 1 - 1/4 = 3/4$$

Answer: 3/4

LEVEL ADVANCED: Prob of A solving = 1/2, B solving = 1/3. What's probability that at least one solves?

Solution: $P(A \text{ solves}) = 1/2$, $P(B \text{ solves}) = 1/3$

$$P(\text{at least one}) = 1 - P(\text{neither solves})$$

$$P(A \text{ doesn't solve}) = 1/2, P(B \text{ doesn't solve}) = 2/3$$

$$P(\text{neither}) = (1/2) \times (2/3) = 1/3$$

$$P(\text{at least one}) = 1 - 1/3 = 2/3$$

Answer: 2/3

F. Common Mistakes

- ✗ Assuming all outcomes equally likely (they're not always)
- ✗ Using addition rule without subtracting overlap
- ✗ Multiplying for mutually exclusive events

G. Exam Tips

- 📍 PLACEMENT: Basic coin/dice/card problems
- 📍 CAT: Complex multi-event scenarios
- 📍 UPSC: Less common, statistics-based problems

H. Practice Questions

1. Two dice rolled. Probability sum > 10?
2. A bag has 5 red, 3 blue, 2 green balls. Probability of drawing red or blue?

1 6

MIXTURE & ALLIGATION

A. What is This Topic?

Mixing substances with different properties and finding optimal blending ratios—uses weighted average.

B. Key Concepts

- **Mixture:** Combining substances with different qualities
- **Alligation Medial:** Finding average quality
- **Alligation Alternate:** Finding quantities for target quality
- **Quality Parameter:** Price, strength, concentration, etc.

C. Important Formulas

$$\text{Average Quality} = \text{Total Value} / \text{Total Quantity}$$

$$\text{Alligation Alternate: } (Q_1/Q_2) = (P_2 - P_{\text{avg}}) : (P_{\text{avg}} - P_1)$$

$$\text{Check: } (Q_1 \times P_1 + Q_2 \times P_2) / (Q_1 + Q_2) = \text{Average}$$

D. Shortcuts & Tricks

- 💧 Use cross-multiplication: differences are proportional
- 💧 Ratio inversely proportional to distance from average
- 💧 Always verify answer by checking average

E. Solved Examples

LEVEL BASIC: 5L of 30% solution mixed with 3L of 50% solution. Average concentration?

Solution: Total value = $(5 \times 30 + 3 \times 50) = 150 + 150 = 300$

Total quantity = $5 + 3 = 8\text{L}$

Average = $300/8 = 37.5\%$

Answer: 37.5%

LEVEL INTERMEDIATE: Milk costs Rs 40/L, water is free. Mix to get Rs 30/L average. Ratio?

Solution: Using alligation: 40 at distance $|40-30| = 10$, 0 at distance $|0-30| = 30$

Ratio of milk:water = $30:10 = 3:1$

Answer: 3:1

LEVEL ADVANCED: Mix Rs 50 and Rs 70 per kg salt to get Rs 65/kg average. Quantities for 100kg?

Solution: Distances: $|70-65| = 5$, $|65-50| = 15$

Ratio = $15:5 = 3:1$

Parts = 4 total. For 100kg: $(3/4) \times 100 = 75\text{kg}$ @ Rs 70, $(1/4) \times 100 = 25\text{kg}$ @ Rs 50

Answer: 75kg @ Rs70, 25kg @ Rs50

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PROBLEMS ON TRAINS

A. What is This Topic?

Speed and distance problems specific to trains—crossing platforms, signals, other trains with relative speed.

B. Key Concepts

- **Train Length:** Length of the train itself
- **Platform/Bridge:** Objects with their own length
- **Signal/Point:** Objects with negligible length
- **Crossing:** Entire length of train must pass the object

C. Important Formulas

Train crosses signal: Distance = Train length, Time = Train length / Speed

Train crosses platform: Distance = Train length + Platform length

Two trains cross: Distance = Train1 length + Train2 length, Relative speed used

Speed = Distance / Time | Time = Distance / Speed

D. Shortcuts & Tricks

- 💧 Train crossing signal: Use ONLY train length
- 💧 Train crossing platform/bridge: Add train length + platform/bridge length
- 💧 Two trains: Use relative speed for approaching, resultant for crossing
- 💧 Remember: km/h to m/s multiply by 5/18, reverse multiply by 18/5

E. Solved Examples

LEVEL BASIC: Train 120m long crosses signal in 6 seconds. Speed?

Solution: Distance = Train length = 120m

Speed = Distance/Time = $120/6 = 20$ m/s

In km/h: $20 \times 18/5 = 72$ km/h

Answer: 72 km/h

LEVEL INTERMEDIATE: 100m train crosses 150m platform in 10 seconds. Speed?

Solution: Total distance = Train + Platform = $100 + 150 = 250$ m

Speed = $250/10 = 25$ m/s = $25 \times 18/5 = 90$ km/h

Answer: 90 km/h

LEVEL ADVANCED: Two trains 80m and 100m, speeds 60 and 40 km/h.

Time to cross each other?

Solution: Total length = $80 + 100 = 180\text{m}$

Relative speed = $60 + 40 = 100\text{ km/h} = 100 \times \frac{5}{18} = 27.78\text{ m/s}$

Time = $180/27.78 \approx 6.48\text{ seconds}$

Answer: $\approx 6.48\text{ seconds}$

F. Common Mistakes

- ✗ Forgetting to add platform length when train crosses platform
- ✗ Using wrong relative speed for approaching vs parallel trains
- ✗ Unit conversion errors ($\text{km/h} \leftrightarrow \text{m/s}$)

G. Exam Tips

- 📍 PLACEMENT: Basic train crossing signal/platform problems
- 📍 CAT: Complex multi-train scenarios
- 📍 UPSC: May appear in transportation contexts

H. Practice Questions

1. Train 150m crosses a bridge of 250m in 20 seconds. Speed in km/h?
2. Train A (100m, 60 km/h) and Train B (120m, 40 km/h) approaching. Time to cross?

1 8 HEIGHTS & DISTANCES (TRIGONOMETRY BASICS)

A. What is This Topic?

Using trigonometry (angles of elevation/depression) to find heights and distances —practical geometry.

B. Key Concepts

- **Angle of Elevation:** Angle looking UP from horizontal
- **Angle of Depression:** Angle looking DOWN from horizontal
- **Horizontal Distance:** Ground distance between observer and object
- **Height:** Vertical distance from ground to top

C. Important Trigonometric Ratios

$$\sin(\theta) = \text{Opposite} / \text{Hypotenuse} \mid \cos(\theta) = \text{Adjacent} / \text{Hypotenuse} \mid \tan(\theta) = \text{Opposite} / \text{Adjacent}$$

$$\text{Height} = \text{Distance} \times \tan(\text{angle}) \mid \text{Distance} = \text{Height} / \tan(\text{angle})$$

D. Common Angles

Angle	30°	45°	60°	90°
sin	1/2	1/√2	√3/2	1
cos	√3/2	1/√2	1/2	0
tan	1/√3	1	√3	∞

E. Solved Examples

LEVEL BASIC: Observer at distance 100m sees top of tower at 45° elevation. Height?

Solution: Height = Distance × tan(45°)
= 100 × tan(45°) = 100 × 1 = 100m

Answer: 100m

LEVEL INTERMEDIATE: Tower height 100m. Observer at angle of elevation 30°. Distance from tower?

Solution: Height = Distance $\times$ tan(angle)

$$100 = \text{Distance} \times \tan(30^\circ)$$

$$100 = \text{Distance} \times (1/\sqrt{3})$$

$$\text{Distance} = 100\sqrt{3} \approx 173.2\text{m}$$

Answer: $100\sqrt{3} \approx 173.2\text{m}$

LEVEL ADVANCED: From cliff top (height 100m), angle of depression to boat is 30°. Distance from cliff?

Solution: Angle of depression = Angle of elevation (from boat's perspective)

$$\text{Height} = \text{Distance} \times \tan(\text{angle})$$

$$100 = \text{Distance} \times \tan(30^\circ) = \text{Distance} \times (1/\sqrt{3})$$

$$\text{Distance} = 100\sqrt{3} \approx 173.2\text{m}$$

Answer: $100\sqrt{3} \approx 173.2\text{m}$

F. Common Mistakes

- ✗ Confusing angle of elevation with angle of depression
- ✗ Using sin when tan needed (or vice versa)
- ✗ Forgetting to rationalize denominators with $\sqrt{}$ values

G. Exam Tips

- 📍 PLACEMENT: Simple height/distance with standard angles (30°, 45°, 60°)
- 📍 CAT: Complex scenarios, multiple observers, moving objects
- 📍 UPSC: Less common, may appear in surveying contexts

H. Practice Questions

1. From distance 50m, angle of elevation is 60°. Height of tree?

2. Two observers 100m apart see building at elevations 30° and 60° . Height?

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DATA INTERPRETATION

A. What is This Topic?

Analyzing data from tables, graphs, pie charts, and caselets to answer questions—key for CAT and UPSC.

B. Types of Data Representation

- **Tables:** Rows and columns of numerical data
- **Bar Charts:** Rectangular bars for comparison
- **Line Graphs:** Points connected by lines, trends
- **Pie Charts:** Circular division by percentage/proportion
- **Caselets:** Passage-based data, requires reading comprehension

C. Key Skills

Read axis labels carefully | Identify units (% , actual numbers, thousands, millions)

For pie charts: $\text{Angle} = (\text{Percentage}/100) \times 360^\circ$ | $\text{Percentage} = (\text{Angle}/360^\circ) \times 100$

$\text{Growth Rate} = ((\text{New} - \text{Old})/\text{Old}) \times 100$ | $\text{Average} = \text{Sum}/\text{Count}$

D. Shortcuts & Tricks

💧 Always check axis labels and units first

💧 For pie charts: $360^\circ = 100\%$, so $1\% \approx 3.6^\circ$

💧 Identify trends (increasing/decreasing) before calculating

💧 For complex caselets, make quick notes of data

E. Solved Example (Table)

LEVEL BASIC: Table shows sales (in 1000s): A=20, B=30, C=25. Total?

Solution: Total = $20 + 30 + 25 = 75$

In actual sales = $75 \times 1000 = 75,000$

Answer: 75,000

LEVEL INTERMEDIATE: Pie chart shows A=40%, B=35%, C=25% of 500.

How many in A?

Solution: A's portion = 40% of 500

= $0.4 \times 500 = 200$

Answer: 200

LEVEL ADVANCED: Line graph shows growth from 100 (year 1) to 150 (year 3). Growth rate?

Solution: Growth = $150 - 100 = 50$

Total growth rate = $(50/100) \times 100 = 50\%$

Annual growth $\approx \sqrt{(150/100)} - 1 \approx 0.225$ or 22.5% per year

Answer: Total 50%, Annual $\approx$ 22.5%

F. Common Mistakes

✗ Misreading axis labels or units

✗ Confusing percentages with actual values

✗ Not reading all data points before concluding

G. Exam Tips

📍 PLACEMENT: Simple table/chart reading with direct questions

📍 CAT: Complex multi-data interpretation, comparative analysis

📍 UPSC: Caselets with social/economic data, require calculation and inference

H. Practice Strategy

1. First, identify the type of data and axis labels
2. Make quick notes of all values
3. Read questions carefully—some ask for ranges, others for exact values
4. Use approximation when exact values not needed
5. Check reasonableness of answers

20 ALGEBRA BASICS

A. What is This Topic?

Solving equations and inequalities—foundation for all advanced aptitude topics.

B. Key Concepts

- **Linear Equation:** Degree 1 ($ax + b = 0$)
- **Quadratic Equation:** Degree 2 ($ax^2 + bx + c = 0$)
- **Inequality:** Comparison using $>$, $<$, $\geq$, $\leq$
- **Solution:** Value(s) that satisfy the equation

C. Important Formulas

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Sum of roots: $-b/a$ | Product of roots: c/a

Discriminant: $D = b^2 - 4ac$ (determines root nature)

D. Shortcuts & Tricks

- 💧 For factoring: Look for factors of 'ac' that sum to 'b'
- 💧 If $D > 0$: Two real roots | $D = 0$: One root | $D < 0$: No real roots
- 💧 For inequalities: Flip sign when multiplying by negative

E. Solved Examples

LEVEL BASIC: Solve $2x + 5 = 13$

Solution: $2x = 13 - 5 = 8$

$$x = 8/2 = 4$$

Answer: $x = 4$

LEVEL INTERMEDIATE: Solve $x^2 - 5x + 6 = 0$

Solution: Factors: $(x - 2)(x - 3) = 0$

$$x = 2 \text{ or } x = 3$$

OR using formula: $x = (5 \pm \sqrt{(25-24)})/2 = (5 \pm 1)/2$

$$x = 3 \text{ or } x = 2$$

Answer: $x = 2$ or $x = 3$

LEVEL ADVANCED: Solve $3x^2 + 7x + 2 = 0$

Solution: $a=3, b=7, c=2$

$$D = 49 - 24 = 25$$

$$x = (-7 \pm 5)/6$$

$$x = -2/6 = -1/3 \text{ or } x = -12/6 = -2$$

Answer: $x = -1/3$ or $x = -2$

F. Common Mistakes

- ✗ Forgetting to flip inequality sign when multiplying by negative
- ✗ Arithmetic errors in quadratic formula

✗ Not checking if solutions satisfy original equation

G. Exam Tips

- 📍 PLACEMENT: Simple linear equations and basic quadratics
- 📍 CAT: System of equations, complex inequalities
- 📍 UPSC: Applied algebra problems in real contexts

H. Practice Questions

1. Solve: $4x - 8 = 20$
2. Solve: $x^2 - 7x + 12 = 0$
3. Solve: $2x^2 + 5x - 3 = 0$

2 1 GEOMETRY & 2D SHAPES

A. What is This Topic?

Properties and calculations for triangles, circles, and polygons—fundamental spatial reasoning.

B. Key Concepts

- **Triangle:** Sum of angles = 180° , types: equilateral, isosceles, scalene
- **Circle:** All points equidistant from center, circumference = $2\pi r$, area = πr^2
- **Polygon:** Sum of angles = $(n-2) \times 180^\circ$
- **Parallel Lines:** Alternate angles equal, corresponding angles equal

C. Important Formulas

$$\text{Triangle: Area} = (1/2) \times \text{base} \times \text{height}$$

$$\text{Circle: Circumference} = 2\pi r, \text{ Area} = \pi r^2, \text{ Arc length} = (\theta/360^\circ) \times 2\pi r$$

$$\text{Pythagorean Theorem: } a^2 + b^2 = c^2 \text{ (right triangle)}$$

$$\text{Polygon area: } (1/2) \times \text{perimeter} \times \text{apothem}$$

D. Key Properties

Shape	Perimeter	Area
Square (side a)	$4a$	a^2
Rectangle (l×w)	$2(l+w)$	$l \times w$
Triangle (sides a,b,c)	$a+b+c$	$(1/2) \times \text{base} \times \text{height}$
Circle (radius r)	$2\pi r$	πr^2

E. Solved Examples

LEVEL BASIC: Area of rectangle 8m × 5m?

Solution: Area = length × width = $8 \times 5 = 40 \text{ m}^2$

Answer: 40 m^2

LEVEL INTERMEDIATE: Circle radius 7cm. Circumference and area?

Solution: Circumference = $2\pi r = 2 \times \pi \times 7 = 14\pi \text{ cm} \approx 44 \text{ cm}$

Area = $\pi r^2 = \pi \times 7^2 = 49\pi \text{ cm}^2 \approx 154 \text{ cm}^2$

Answer: Circumference $\approx 44\text{cm}$, Area $\approx 154\text{cm}^2$

LEVEL ADVANCED: Right triangle sides 3, 4, 5. Inradius?

Solution: Area = $(1/2) \times 3 \times 4 = 6$

Semi-perimeter = $(3+4+5)/2 = 6$

Inradius = Area/Semi-perimeter = $6/6 = 1$

Answer: 1

F. Common Mistakes

- ✗ Confusing radius with diameter
- ✗ Forgetting angle sum properties
- ✗ Using diameter instead of radius in circle formulas

G. Exam Tips

- 📍 PLACEMENT: Simple area/perimeter calculations
- 📍 CAT: Complex composite shapes, overlapping regions
- 📍 UPSC: Geometric reasoning and proofs

H. Practice Questions

1. Square with side 6cm. Diagonal length?
2. Circle radius 10cm. Sector area with central angle 60° ?

2 2 MENSURATION (3D SHAPES)

A. What is This Topic?

Volume, surface area, and calculations for 3D shapes—cubes, cylinders, spheres, cones.

B. Key Concepts

- **Volume:** Space occupied by 3D shape
- **Surface Area:** Total outer area of 3D shape

- **Lateral Surface Area:** Area excluding top and bottom
- **Cube/Cuboid:** Rectangular 3D shapes

C. Important Formulas

$$\text{Cube: Volume} = a^3, \text{ Surface Area} = 6a^2, \text{ Lateral} = 4a^2$$

$$\text{Cylinder: Volume} = \pi r^2 h, \text{ Lateral SA} = 2\pi r h, \text{ Total SA} = 2\pi r(r+h)$$

$$\text{Sphere: Volume} = (4/3)\pi r^3, \text{ Surface Area} = 4\pi r^2$$

$$\text{Cone: Volume} = (1/3)\pi r^2 h, \text{ Lateral SA} = \pi r l, \text{ Total SA} = \pi r(r+l)$$

D. Shortcuts & Tricks

- 💧 For sphere: Volume is $2/3$ of circumscribed cylinder
- 💧 For cone: Volume is $1/3$ of cylinder with same base and height
- 💧 Slant height of cone: $l = \sqrt{r^2 + h^2}$

E. Solved Examples

LEVEL BASIC: Cube side 5cm. Volume and surface area?

Solution: Volume = $5^3 = 125 \text{ cm}^3$

Surface Area = $6 \times 5^2 = 6 \times 25 = 150 \text{ cm}^2$

Answer: Volume = 125cm^3 , SA = 150cm^2

LEVEL INTERMEDIATE: Cylinder $r=7\text{cm}$, $h=10\text{cm}$. Volume?

Solution: Volume = $\pi r^2 h = \pi \times 7^2 \times 10 = 490\pi \approx 1540 \text{ cm}^3$

Answer: $\approx 1540 \text{ cm}^3$

LEVEL ADVANCED: Sphere radius 6cm. Volume and surface area?

Solution: Volume = $(4/3)\pi r^3 = (4/3)\pi \times 6^3 = (4/3)\pi \times 216 = 288\pi \approx 905 \text{ cm}^3$

Surface Area = $4\pi r^2 = 4\pi \times 36 = 144\pi \approx 452 \text{ cm}^2$

Answer: Volume $\approx 905\text{cm}^3$, SA $\approx 452\text{cm}^2$

F. Common Mistakes

- ✗ Confusing volume and surface area formulas
- ✗ Using diameter instead of radius
- ✗ Forgetting π or using wrong value

G. Exam Tips

- 📍 PLACEMENT: Direct volume/SA calculations
- 📍 CAT: Complex composite 3D shapes
- 📍 UPSC: Applied mensuration in real contexts

H. Practice Questions

1. Cuboid $6 \times 4 \times 3\text{cm}$. Volume and surface area?
2. Cone $r=5\text{cm}$, $h=12\text{cm}$. Slant height and volume?

ALL THE BEST